individual-assignment5

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2026-05-08

Set Up

Packages

```r
# general use and cleaning
library(tidyverse)
library(janitor)
# file organization
library(here)
# wrapping long axis labels
library(scales)
# reading in .xlsx files
library(readxl)
# visualizing missing data
library(naniar)
# arranging multiple plots
library(patchwork)
# converting to snake case
library(snakecase)
```

Data

```r
# taxonomic information for each taxon ID
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrate count data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```
                    sheet = "Aquatic Insects")

# site metadata
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

# water quality measurements
water_quality <- read_xlsx(here("data", "Aquatic Sampling
↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Class Code

```
# filter sites to only quarterly NCOS sites for downstream filtering joins
ncos_sites <- sites |>
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```
# convert taxon names to snake_case so they match column names in aq_ins
taxon_list_clean <- taxon_list |>
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

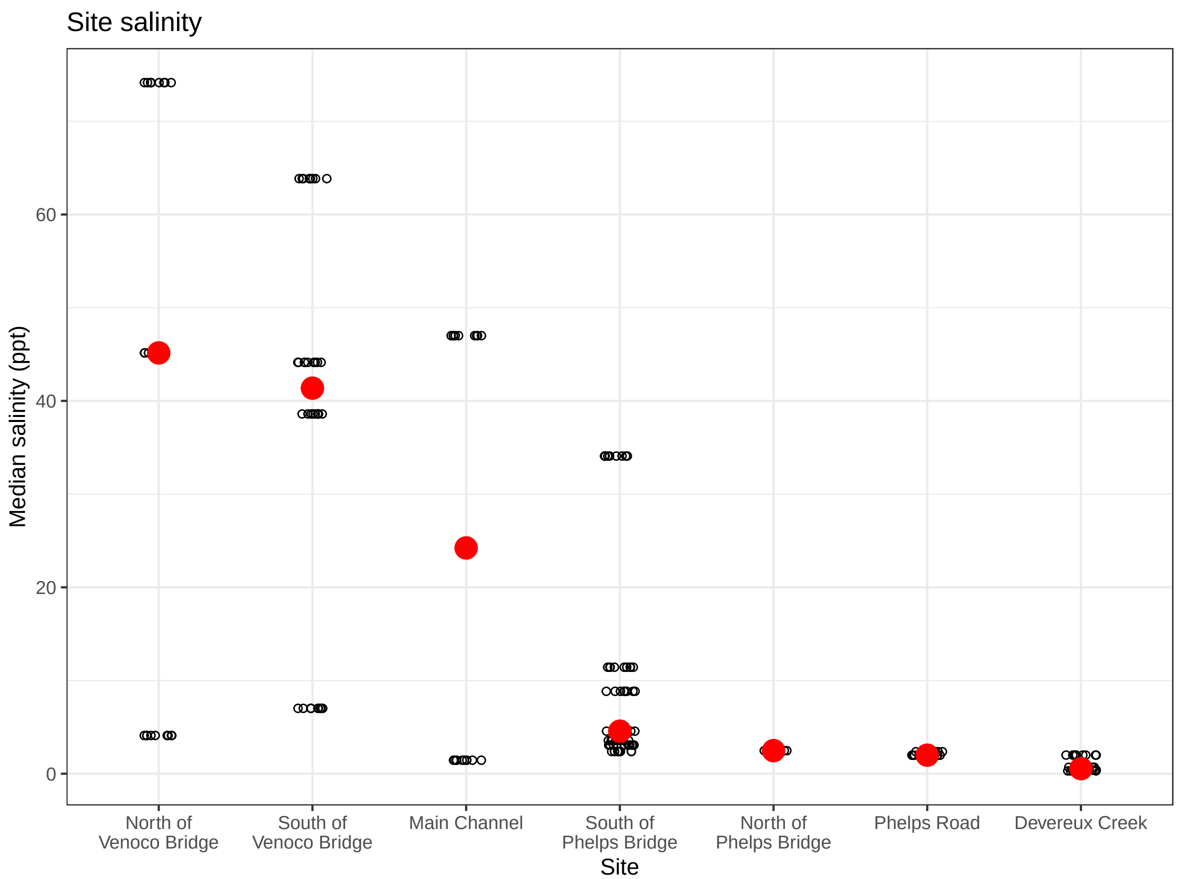
```
# create summarized water quality data frame for each date-site combination
water_quality_clean <- water_quality |>
  # standardize column names
  clean_names() |>
  # keep only NCOS quarterly sites
  semi_join(ncos_sites, by = "site") |>
  # create a single date_site key for joining
  unite("date_site", date, site, remove = TRUE) |>
  # group by survey event
  group_by(date_site) |>
  # calculate median water quality metrics per survey event
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # remove a known salinity outlier
```

```
  filter(date_site != "2022-08-12_NVBR")
```

```
# create main analysis data frame: long format with one row per
# taxon per survey event, with water quality and taxonomy joined in
aq_ins_clean <- aq_ins |>
  # standardize column names
  clean_names() |>
  # keep only NCOS quarterly sites
  semi_join(ncos_sites, by = c("site")) |>
  # rename date column for clarity
  rename(date = date_on_vial) |>
  # select only relevant taxon and metadata columns
  select(date, sample_type, site,
         ostracod, copepod,
         hemiptera_corixidae_boatman,
         cladocera, nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  # keep only filtered beaker samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  # pivot to long format so each row is one taxon per survey
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  # group by survey event and taxon
  group_by(date, site, taxon) |>
  # calculate average abundance per liter (total count / 7.5 L)
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  # ungroup data frame
  ungroup() |>
  # create date_site key for joining with water quality
  unite("date_site", date, site, remove = FALSE) |>
  # attach water quality measurements
  left_join(water_quality_clean, by = "date_site") |>
  # attach full taxonomic information
  left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
  # add human-readable site names
  mutate(site_full = case_when(
    site == "NVBR"  ~ "North of Venoco Bridge",
    site == "NEC"   ~ "South of Venoco Bridge",
```

```r
    site == "NMC"   ~ "Main Channel",
    site == "NPB"   ~ "South of Phelps Bridge",
    site == "NPB1"  ~ "North of Phelps Bridge",
    site == "NPB2"  ~ "Phelps Road",
    site == "NWP"   ~ "West Pond",
    site == "NDC"   ~ "Devereux Creek"
  )) |>
  # order sites by median salinity (high to low) for consistent axis ordering
  mutate(site_full = fct_reorder(site_full, med_sal,
                                .fun = "median", .na_rm = TRUE),
         site_full = fct_rev(site_full))
```

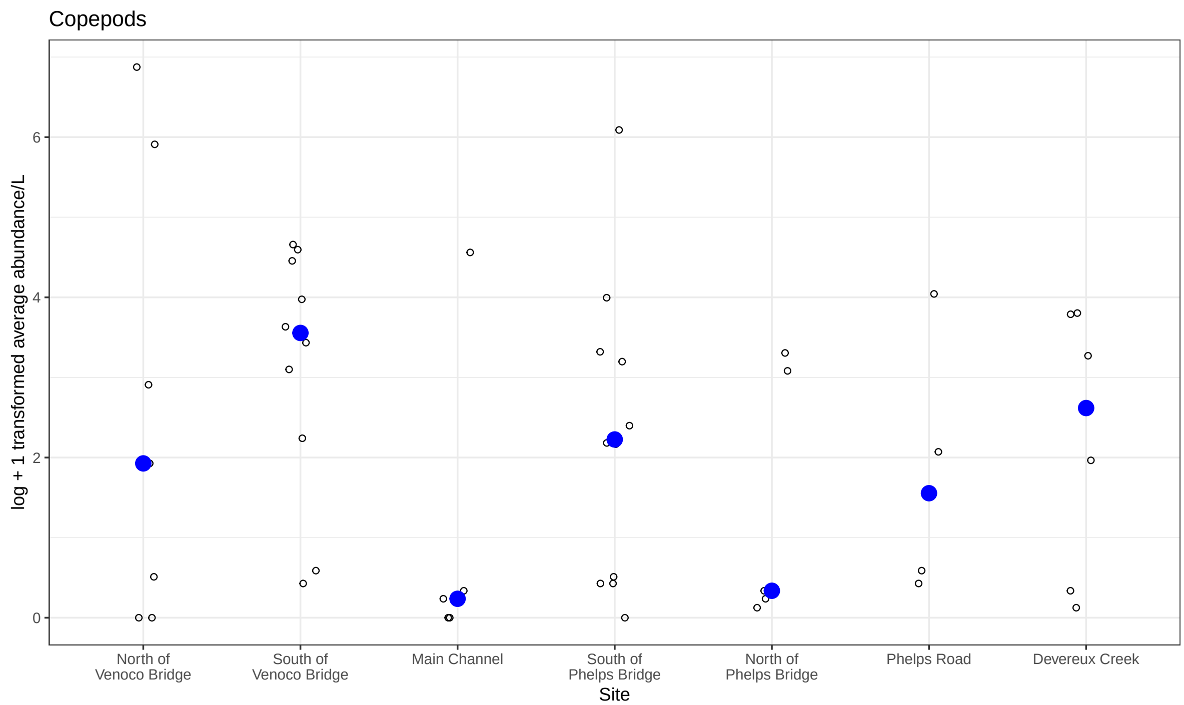
```r
# salinity panel: one point per survey, median shown in red
salinity <- ggplot(data = aq_ins_clean,
                   mapping = aes(x = site_full, y = med_sal)) +
  # individual survey salinity measurements
  geom_jitter(height = 0, width = 0.1, shape = 21) +
  # wrap long site names on x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # red point for median salinity per site
  stat_summary(fun = median, color = "red", size = 1) +
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  theme_bw()

salinity
```



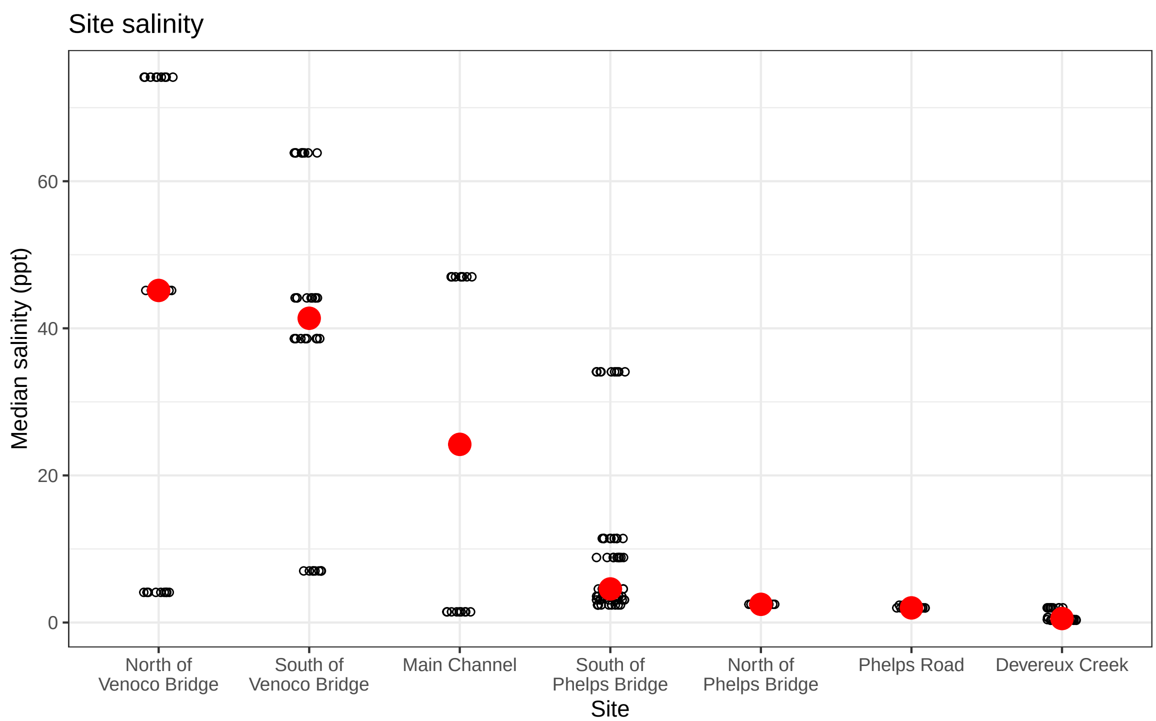

```
# filter to copepods only and log+1 transform abundance
copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  mutate(log_ave_lit = log(ave_lit + 1))
```

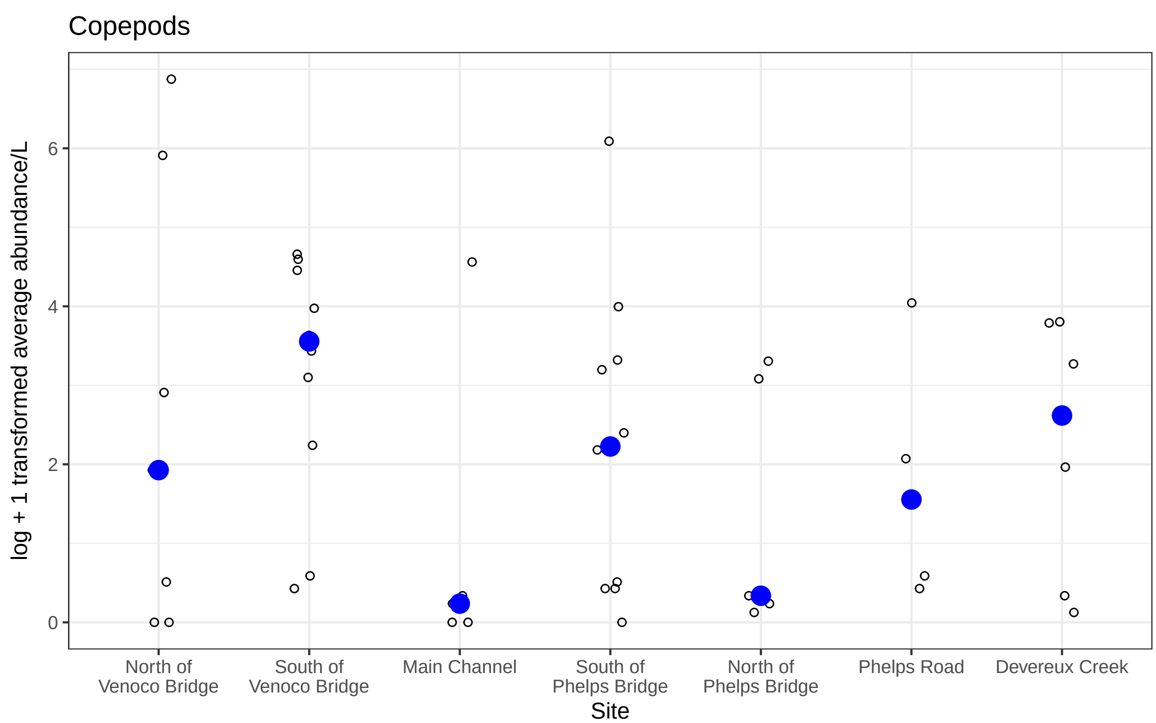
```
# copepod panel: individual observations + blue median point
copepods_plot <- ggplot(data = copepods,
                        mapping = aes(x = site_full, y = log_ave_lit)) +
  # individual observations as open circles
  geom_jitter(height = 0, shape = 21, width = 0.1) +
  # blue point for median per site
  stat_summary(geom = "point", fun = "median",
               size = 4, color = "blue") +
  scale_x_discrete(labels = label_wrap(15)) +
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
```

```r
      title = "Copepods") +
  theme_bw()

copepods_plot
```




```r
# two-panel figure from class: salinity stacked above copepods
salinity / copepods_plot
```


Site salinity
Median salinity (ppt)
0
20
40
60
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site


Copepods
log + 1 transformed average abundance/L
0
2
4
6
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site

a. Planning the figure

- x axis: `site_full` (site name)
- y axis: log + 1 transformed average ostracod abundance per liter
- geometry to represent average abundance/L: `geom_jitter()` to plot every individual survey observation as points
- geometry to represent medians: stat_summary(fun = "median") overlays colored points to show medians per site

b. Wragling Data

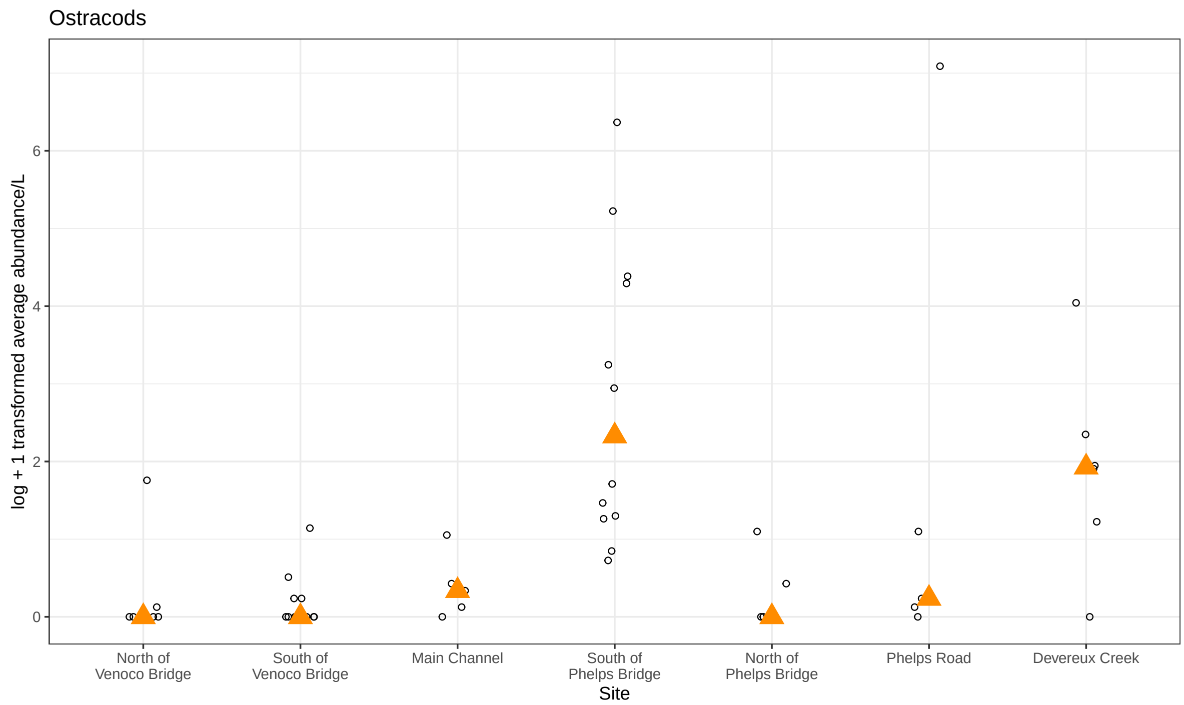
- filter `aq_ins_clean` to only include ostracods
- log +1 transform average abundances

```
# creating new object from aq_ins_clean
ostracods <- aq_ins_clean |>
  # filter to only include the taxon titled ostracod
  filter(taxon == "ostracod") |>
  # creates new column called log_ave_lit that contains the log + 1
  ↪ transformed average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2022-08-10_NPB  2022-08-10 00:00:00 NPB   ostr~   2.67    8.28   8.84    9.81
 2 2022-08-12_NEC  2022-08-12 00:00:00 NEC   ostr~   0       7.37  NA      NA
 3 2023-11-14_NPB  2023-11-14 00:00:00 NPB   ostr~  18      NA     NA      NA
 4 2022-02-23_NVBR 2022-02-23 00:00:00 NVBR  ostr~   0       8.86  45.2    10.2
 5 2021-11-23_NPB2 2021-11-23 00:00:00 NPB2  ostr~   0.267   7.59   1.99    0.65
 6 2022-05-06_NMC  2022-05-06 00:00:00 NMC   ostr~   0       9.96   1.45   16.5
 7 2023-05-04_NMC  2023-05-04 00:00:00 NMC   ostr~   1.87    8.32  NA      10.3
 8 2023-01-28_NPB1 2023-01-28 00:00:00 NPB1  ostr~   0       7.66  NA       9.74
 9 2023-03-03_NDC  2023-03-03 00:00:00 NDC   ostr~   5.73    7.45   0.688   5.64
10 2022-05-05_NEC  2022-05-05 00:00:00 NEC   ostr~   0.267   9.45  63.8     7.3
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Coding the figure

```
# base layer: x = site, y = log transformed ostracod abundance
ostracods_plot <- ggplot(data = ostracods,
                         mapping = aes(x = site_full,
                                       y = log_ave_lit)) +
  # individual survey observations as open circles
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  # median per site: large filled orange triangle to stand out from
   ↪  observations
  stat_summary(geom = "point",
               fun = "median",
               size = 5,
               color = "darkorange",
               shape = 17) +
  # wrap long site names so they don't overlap on x-axis
  scale_x_discrete(labels = label_wrap(15)) +
  # clear axis labels and title naming the taxonomic group
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracods") +
  # clean theme different from default gray
  theme_bw()

# display the plot
ostracods_plot
```


Ostracods
log + 1 transformed average abundance/L
0
2
4
6
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site